

$$\frac{d}{dx} c = 0$$

Constant function

$$\frac{d}{dx} x^n = n x^{n-1}$$

Power rule - n any real
number. [Binomial theorem

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Examples:

limit definition!!

No proof.

$$1. \frac{d}{dx} x^e = e x^{e-1}$$

$$2. \frac{d}{dx} \sqrt[3]{x^4} = \frac{d}{dx} x^{\frac{4}{3}} = \frac{4}{3} x^{\frac{1}{3}}$$

$$3.1 \frac{d}{dx} x \sqrt[3]{x} = \frac{d}{dx} x \cdot x^{\frac{1}{3}} = \frac{d}{dx} x^{\frac{4}{3}} = \frac{4}{3} x^{\frac{1}{3}}$$

3.2. Find the slope of the tangent and
normal line at (1, 1)

$$x=1 \quad f'(1) = \frac{4}{3} \quad \text{Slope}$$

tang.

$$\left[\begin{array}{l} y - y_1 = m(x - x_1) \\ \text{Eq. tang.} \end{array} \right]$$

normal line slope $-\frac{3}{4}$

$$\frac{d}{dx} [c f(x)] = c \frac{d}{dx} f(x) * \text{Constant multiple rule}$$

$$\frac{d}{dx} [f(x) + g(x)] = \frac{d}{dx} f(x) + \frac{d}{dx} g(x) \quad \text{Sum rule}$$

$$\frac{d}{dx} [f(x) - g(x)] = \frac{d}{dx} f(x) - \frac{d}{dx} g(x) \quad \text{Difference rule}$$

$$\frac{d}{dx} (2x^2) = 2 \cdot 2x = 4x$$

$$\frac{d}{dx} (2x^2 + x^3) = 4x + 3x^2$$

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Find $\frac{d}{dx} e^x$ using first principles.

$$\begin{aligned} \frac{d}{dx} e^x &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \rightarrow e^x \cdot e^h \\ &= \lim_{h \rightarrow 0} \frac{e^{x+h} - e^x}{h} \quad \left(\frac{0}{0} \right) \\ &= \lim_{h \rightarrow 0} e^x \frac{(e^h - 1)}{h} \\ &= e^x \boxed{\lim_{h \rightarrow 0} \frac{(e^h - 1)}{h}} = e^x \rightarrow \end{aligned}$$

The number $e = 2.7182\ldots$ satisfies

$$\lim_{h \rightarrow 0} \frac{(e^h - 1)}{h} = 1 \quad ! ! ! !$$

$$\frac{d}{dx} e^x = e^x$$

Examples: Find $\frac{dy}{dx}$ for each of the following.

1. $y = 12$

$$y' = 0$$

2. $y = x^5$

$$y' = 5x^4$$

3. $y = 2x^4$

$$y' = 8x^3$$

4. $y = x^3 + 5$

$$y' = 3x^2$$

5. $y = 3x^2 - 2x + \pi$

$$y' = 6x - 2$$

6. $y = x + \frac{2}{x} = x + 2x^{-1} \Rightarrow \frac{dy}{dx} = 1 - \frac{2}{x^2}$

7. $y = \frac{x}{2} - \sqrt{x} \quad y' = \frac{1}{2} - \frac{1}{2\sqrt{x}}$

8. $y = 3x^2 + e$

$$y' = 6x$$

9. $y = e^{3+x} + x^2 = e^3 \cdot e^x + x^2 \Rightarrow \frac{dy}{dx} = e^3 \cdot e^x + 2x$

10. $y = e^x + \frac{1}{x}$

$$\therefore y' = e^x - \frac{1}{x^2}$$

11. Find the x -value(s) where you can draw a horizontal tangent to

$$f(x) = (x-4)(x+3) \quad [f'(x) = 0]$$

$$= x^2 - x - 12$$

$$\Rightarrow f'(x) = 2x - 1 = 0 \text{ at horizontal tangent}$$

So $2x - 1 = 0 \Rightarrow x = \frac{1}{2}$. There is a horizontal tangent at $x = \frac{1}{2}$.

12. Find the equation of the tangent to $f(x) = 3e^x - \frac{4}{x}$ where $x = 2$.

$$f(2) = 3e^2 - 2$$

$$\text{and } f'(x) = 3e^x + \frac{4}{x^2} \text{ so } f'(2) = 3e^2 + 1$$

Eq. of the tangent is:

$$y - (3e^2 - 2) = (3e^2 + 1)(x - 2)$$

$$\Rightarrow y = (3e^2 + 1)x - 3e^2 - 4$$